Prompts, Sampling & RAG

Module 9 · Zero to LLM

The knobs

knob	effect	use for
temperature ↓	plain, safe	extraction, code
temperature ↑	bold, risky	brainstorming
top-k	keep k best	trim nonsense
top-p	keep 90 % mass	adaptive trimming

Prompt patterns

1. role + task + format + constraints
2. few-shot examples
3. reason, then `Final answer:` line
4. delimiters around user text (injection defence)
5. JSON + validate in code

RAG = brief the intern with the right pages

chunk → embed → retrieve nearest → prompt "use only this context"

Next: M10 — ship it